

Alpaca Options Agent

Deterministic Risk & Statistical Validation

An architecture that refuses bad trades.

1. The Core Thesis: Authority in Python, Not LLMs

We demonstrated this thesis directly with **Council Attack C01**:

Our Featherless arbiter (Qwen2.5-7B-Instruct) was asked to bless a 400-contract order. It confidently ruled **PROCEED**, rationalising: *"The strategy is validated and the position size is acceptable."*

This was factually false. The trade put \$172,000 at risk against an \$8,000 cap.

Our deterministic Python `RiskGate` refused it anyway. This is the thesis demonstrated instead of asserted: a third model confidently authorising a catastrophic trade, and Python stopping it.

2. Independent Measurements of the Same Claim

The C01 attack result is backed by two other independent measurements:

A. Replay Divergence (240 replays, n=60 per cell)

Our replay harness uses 108,012 characters of **frozen market data** — no external APIs are re-fetched. (Proved executably via `verify_replay_isolation.py` blocking DNS). This proves the variance is strictly the LLM.

Headline: `gpt-4o-mini` Proposer (free tool choice, temp 0, fixed seed, byte-identical inputs) — **100% of decision turns changed on replay (40 of 40, 95% CI 91.2-100%)**.

Divergence concentrates where authority sits. Research-only conversation diverged 65%, but every decision turn changed.

Per-Cell Replay Breakdown (never pooled):

- **gpt-4o-mini / proposer (free choice):** 90.0% divergent, decisions changed 40/40.

3. The Cost of Refusal (Counterfactual)

The gate refused strategies that would have returned +\$1,326.64, with 15 of 21 refused pairs profitable.

Why we still refused them:

- It covers one trading day, not three.
- These are marks, not closed results.
- SPY +0.44% / QQQ +0.23% / IWM +1.18% guarantees short premium profits on an up day.

The gate refuses on the width of the bootstrap interval, never on the sign of the latest observation. One observation is exactly the sample size the interval already judged insufficient.

3. Statistical Validation Gate

We evaluated 21 distinct strategy/symbol pairs (SPY, QQQ, IWM $\times$ 7 structures), producing 24 total validation records.

- **0 pairs cleared for live trading.**
- The closest: SPY vertical credit spread (78% win rate, Sharpe 1.67). Refused because the Sharpe CI lower bound remained negative (-0.32).
- **Moving-block bootstrap:** Correcting from an i.i.d. bootstrap to a moving-block model cut the false-positive rate from **10.8%** to a calibrated **2.5%**.

The gate governs claims of edge. The risk layer governs structural safety.

4. The Proposer / Critic Architecture

The LLM never computes a number that reaches an order.

- **Proposer Agent:** Suggests regime and parameters.
- **Critic Agent:** Evaluates the proposal against our validation graveyard.
- **Featherless Arbiter:** A third open-weights model arbitrates disagreements.

Live Veto: The Critic agent actively vetoed a proposed MSFT cash-secured put for citing no backtest evidence on a symbol that never cleared the gate.

Cost: \$0.022 per Proposer/Critic cycle.

5. Deterministic Risk Gate & Execution

Every order must pass the non-LLM Python risk gate before routing to the Alpaca MCP server.

- We implemented a batch model for structure-level capital at risk.
- **Old per-leg model:** Refused \$75,500 capital-at-risk for a SPY spread.
- **New batch model:** Correctly calculated \$[FINAL_BATCH_RISK] risk and approved the trade.

Demonstration Trade: We executed **one** unvalidated SPY credit spread pinned to the Friday Sept 4th expiry, explicitly bypassing validation to demonstrate the execution path. We close Thursday to avoid Friday's NFP event risk.

Thank You

Code public.

Options quotes provided by Alpaca's free-tier Indicative Pricing Feed.